Incidence and predictors of delirium after cardiac surgery: Results from The IPDACS Study.

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Delirium after cardiac surgery is a serious complication that results in higher morbidity and mortality rates, and prolongs hospitalisation. However, the knowledge base regarding the issue of postoperative delirium is still limited. Therefore, in the current study, we evaluated the incidence and independent perioperative risk factors of delirium after cardiac surgery. The IPDACS Study recruited 563 consecutive patients undergoing cardiac surgery with cardiopulmonary bypass. The subjects were preoperatively examined by psychiatrists using the Mini-Mental State Examination and the Mini International Neuropsychiatric Interview to assess psychiatric comorbidity. Additionally, other variables connected to the patients' medical condition and surgical and anaesthetic procedures were evaluated. A diagnosis of delirium following surgical intervention was based on Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) criteria. The incidence of postoperative delirium according to DSM-IV criteria was 16.3% (95% confidence interval: 13.5-19.6). Multivariate stepwise logistic regression analysis revealed that advanced age, preoperative cognitive impairment, an ongoing episode of major depression, anaemia, atrial fibrillation, prolonged intubation and postoperative hypoxia were independently associated with delirium after cardiac surgery. According to the current analysis, the aforementioned conditions independently predispose to delirium following cardiac surgery. Since some of these factors can be successfully treated and eliminated preoperatively and postoperatively, this study should be helpful in reducing the risk of delirium and in improving the medical care of patients undergoing cardiac surgery (Clinical Trials Identifier: NCT00784576).

Identification of risk factors for delirium, cognitive decline, and dementia after cardiac surgery (FINDERI-find delirium risk factors): a study protocol of a prospective observational study.

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Postoperative delirium is a common complication of cardiac surgery associated with higher morbidity, longer hospital stay, risk of cognitive decline, dementia, and mortality. Geriatric patients, patients undergoing cardiac surgery, and intensive care patients are at a high risk of developing postoperative delirium. Gold standard assessments or biomarkers to predict risk factors for delirium, cognitive decline, and dementia in patients undergoing cardiac surgery are not yet available. The FINDERI trial (FINd DElirium RIsk factors) is a prospective, single-center, observational study. In total, 500 patients aged ≥ 50 years undergoing cardiac surgery at the Department of Cardiovascular and Thoracic Surgery of the University of Göttingen Medical Center will be recruited. Our primary aim is to validate a delirium risk assessment in context of cardiac surgery. Our secondary aims are to identify specific preoperative and perioperative factors associated with delirium, cognitive decline, and accelerated dementia after cardiac surgery, and to identify blood-based biomarkers that predict the incidence of postoperative delirium, cognitive decline, or dementia in patients undergoing cardiac surgery. This prospective, observational study might help to identify patients at high risk for delirium prior to cardiac surgery, and to identify important biological mechanisms by which cardiac surgery is associated with delirium. The predictive value of a delirium screening questionnaire in cardiac surgery might be revealed. Finally, the identification of specific blood biomarkers might help to predict delirium, cognitive decline, and dementia in patients undergoing cardiac surgery. Ethics approval for this study was obtained from the IRB of the

University of Göttingen Medical Center. The investigators registered this study in the German Clinical Trials Register (DRKS; https://www.drks.de) (DRKS00025095) on April 19th, 2021.

Association of electroencephalogram epileptiform discharges during cardiac surgery with postoperative delirium: An observational study.

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Delirium is a frequent and serious complication following cardiac surgery involving cardiopulmonary bypass (CPB). Electroencephalography reflects the electrical activity of the cerebral cortex. The impact of electroencephalographic epileptiform discharges during cardiac surgery on postoperative delirium remains unclear. This study was designed to investigate the relationship between intraoperative epileptiform discharges and postoperative delirium in patients undergoing cardiac surgery. A total of 76 patients who underwent cardiac surgery under CPB were included. The baseline cognitive status was measured before surgery. Electroencephalograms were monitored continuously from entry into the operating room to the end of surgery. The presence of delirium was assessed through the Confusion Assessment Method or the Confusion Assessment Method for the Intensive Care Unit on the first 3 days after surgery. Univariate and multivariate logistic regression analyses were performed to evaluate the association between epileptiform discharges and delirium. Delirium occurred in 31% of patients and epileptiform discharges were present in 26% of patients in the study. Patients with delirium had a higher incidence of epileptiform discharges (52.63% vs. 13.95%, $P < 0.001$) and longer durations of anesthesia and CPB ($P = 0.023$ and $P = 0.015$, respectively). In addition, patients with delirium had a longer length of hospital stay and a higher incidence of postoperative complications. Multivariate logistic regression analysis showed that age and epileptiform discharges were significantly associated with the incidence of postoperative delirium [odds ratio, 4.75 (1.26-17.92), $P = 0.022$; 5.00 (1.34-18.74), $P = 0.017$, respectively]. Postoperative delirium is significantly related to the occurrence of epileptiform discharges during cardiac surgery.

Hypertension, mitral valve disease, atrial fibrillation and low education level predict delirium and worst outcome after cardiac surgery in older adults.

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Delirium is a common complication after cardiac surgery in older adult patients. However, risk factors and the influence of delirium on patient outcomes are not well established. We aimed to determine the incidence, predisposing and triggering factors of delirium following cardiac surgery. One hundred seventy-three consecutive patients aged ≥60 years were studied. Patients' characteristics and two cognitive function assessment tests were recorded preoperatively. Perioperative variables were blood transfusion, orotracheal intubation time (OIT), renal dysfunction, and hypoxemia. Delirium was assessed using the Confusion Assessment Method for the Intensive Care Unit. The composite outcome consisted of death, infection, and perioperative myocardial infarction until hospital discharge or 30 days after surgery, and for up to 18 months. One hundred six patients (61.27%) were men and the age was 69.5 ± 5.8 years. EuroSCORE II index was 4.06 ± 3.86. Hypertension was present in 75.14%, diabetes in 39.88%, and 30.06% were illiterate. Delirium occurred in 59 patients (34.1%). Education level (OR 0.81, 0.71-0.92), hypertension (OR 2.73, 1.16-6.40), and mitral valve disease (OR 2.93, 1.32-6.50) were independent predisposing factors for delirium, and atrial fibrillation after surgery (OR 2.49, 1.20-5.20) represented the potential triggering factor. Delirium (OR 2.35, 1.20-4.58) and OIT ≥ 900 min (OR 2.50; 1.30-4.80) were independently associated with the composite outcome. In older adult patients submitted to cardiac surgery, delirium is a frequent complication that is associated with worst outcome. Independent risk factors for delirium included education level, hypertension, mitral valve disease, and atrial fibrillation after cardiac surgery.

Delirium in Children After Cardiac Bypass Surgery.

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To describe the incidence of delirium in pediatric patients after cardiac bypass surgery and explore associated risk factors and effect of delirium on in-hospital outcomes. Prospective observational single-center study. Fourteen-bed pediatric cardiothoracic ICU. One hundred ninety-four consecutive admissions following cardiac bypass surgery, 1 day to 21 years old. Subjects were screened for delirium daily using the Cornell Assessment of Pediatric Delirium. Incidence of delirium in this sample was 49%. Delirium most often lasted 1-2 days and developed within the first 1-3 days after surgery. Age less than 2 years, developmental delay, higher Risk Adjustment for Congenital Heart Surgery 1 score, cyanotic disease, and albumin less than three were all independently associated with development of delirium in a multivariable model (all $p < 0.03$). Delirium was an independent predictor of prolonged ICU length of stay, with patients who were ever delirious having a 60% increase in ICU days compared with patients who were never delirious ($p < 0.01$). In our institution, delirium is a frequent problem in children after cardiac bypass surgery, with identifiable risk factors. Our study suggests that cardiac bypass surgery significantly increases children's susceptibility to delirium. This highlights the need for heightened, targeted delirium screening in all pediatric cardiothoracic ICUs to potentially improve outcomes in this vulnerable patient population.

Incidence and assessment of delirium following open cardiac surgery: a systematic review and meta-analysis.

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This systematic review and meta-analysis sought (i) to provide an overview of the incidence of delirium following open cardiac surgery and (ii) to investigate how incidences of delirium are associated with different assessment tools. A systematic search of studies investigating delirium following open cardiac surgery was conducted in Medline (Ovid), EMBASE, PsycINFO, CiNAHL, and the Cochrane Database. Only studies with patients diagnosed or screened with a validated tool were included. Studies published from 2005-2021 were included in the meta-analysis. Of 7126 individual studies retrieved, 106 met the inclusion criteria for the meta-analysis, hereof 31% of high quality. The weighted pooled incidence of delirium following open cardiac surgery across all studies was 23% (95% CI 20-26%), however we found a considerable heterogeneity ($I^2 = 99\%$), which could not be explained by subgroups or further sensitivity analyses. The most commonly applied screening tool for delirium is CAM/CAM-ICU. The lowest estimates of delirium were found by applying the Delirium Observation Scale (incidence 14%, 95% CI 8-20%), and the highest estimates in studies using 'other' screening tools (Organic Brain Symptom Scale, Delirium Symptom Interview) with a pooled incidence of 43% (95% CI 19-66%), however, only two studies applied these. Delirium following open cardiac surgery remains a complication with a high incidence of overall 23%, when applying a validated tool for screening or diagnosis. Nevertheless, this systematic review and meta-analyses highlight the significant inconsistency in current evidence regarding assessment tools and regimens. Prospero CRD42020215519.

Sleep-disordered breathing is a risk factor for delirium after cardiac surgery: a prospective cohort study.

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Delirium is a frequent complication after cardiac surgery. Although various risk factors for postoperative delirium have been identified, the relationship between nocturnal breathing disorders and delirium has not yet been elucidated. This study evaluated the relationship between sleep-disordered breathing (SDB) and postoperative delirium in cardiac surgery patients without a previous diagnosis of obstructive sleep apnea. In this prospective cohort study, 92 patients undergoing elective cardiac surgery with extracorporeal circulation were evaluated for both SDB and postoperative delirium. Polygraphic recordings were used to calculate the apnea-hypopnea index (AHI; mean number of apneas and hypopneas per hour recorded) of all patients preoperatively. Delirium was assessed during the first four postoperative days using the Confusion Assessment Method. Clinical differences between individuals with and without postoperative delirium were determined with univariate analysis. The relationship between postoperative delirium and those covariates that were associated with delirium in univariate analysis was determined by a multivariate logistic regression model. The median overall preoperative AHI was 18.3 (interquartile range, 8.7 to 32.8). Delirium was diagnosed in 44 patients. The median AHI differed significantly between patients with and without postoperative delirium (28 versus 13; $P = 0.001$). A preoperative AHI of 19 or higher was associated with an almost sixfold increased risk of postoperative delirium (odds ratio, 6.4; 95% confidence interval, 2.6 to 15.4; $P < 0.001$). Multivariate logistic regression analysis showed that preoperative AHI, age, smoking, and blood transfusion were independently associated with postoperative delirium. Preoperative SDB (for example, undiagnosed obstructive sleep apnea) were strongly associated with postoperative delirium, and may be a risk factor for postoperative delirium.

Postoperative delirium prediction after cardiac surgery using machine learning models.

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Postoperative delirium (POD) is a common postoperative complication in elderly patients, especially those undergoing cardiac surgery, which seriously affects the short- and long-term prognosis of patients. Early identification of risk factors for the development of POD can help improve the perioperative management of surgical patients. In the present study, five machine learning models were developed to predict patients at high risk of delirium after cardiac surgery and their performance was compared. A total of 367 patients who underwent cardiac surgery were retrospectively included in this study. Using single-factor analysis, 21 risk factors for POD were selected for inclusion in machine learning. The dataset was divided using 10-fold cross-validation for model training and testing. Five machine learning models (random forest (RF), support vector machine (SVM), radial based kernel neural network (RBFNN), K-nearest neighbour (KNN), and Kernel ridge regression (KRR)) were compared using area under the receiver operating characteristic curve (AUC-ROC), accuracy (ACC), sensitivity (SN), specificity (SPE), and Matthews coefficient (MCC). Among 367 patients, 105 patients developed POD, the incidence of delirium was 28.6 %. Among the five ML models, RF had the best performance in ACC (87.99 %), SN (69.27 %), SPE (95.38 %), MCC (70.00 %) and AUC (0.9202), which was far superior to the other four models. Delirium is common in patients after cardiac surgery. This analysis confirms the importance of the computational ML models in predicting the occurrence of delirium after cardiac surgery, especially the outstanding performance of the RF model, which has practical clinical applications for early identification of patients at risk of developing POD.

Risk factors of postoperative delirium after cardiac surgery: a meta-analysis.

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Postoperative delirium is a frequent event after cardiac surgery. This meta-analysis aimed to identify relevant risk factors. In this meta-analysis, all original researches regarding patients undergoing mixed types of cardiac surgery (excluding transcatheter procedures) and postoperative delirium were evaluated for inclusion. On July 28th 2020, we searched PubMed, Embase, Web of Science and Scopus. Data about name of first author, year of publication, inclusion and exclusion criteria, research design, setting, method of delirium assessment, incidence of delirium, odds ratio (OR) and corresponding 95% confidence interval (CI) of risk factors, and other information relevant was collected. OR and 95% CI were used as metrics for summarized results. Random effects model was applied. Fourteen reports were included with a total sample size of 13,286. The incidence of delirium ranged from 4.1 to 54.9%. Eight risk factors were identified including aging, diabetes, preoperative depression, mild cognitive impairment, carotid artery stenosis, NYHA functional class III or IV, time of mechanical ventilation and length of intensive care unit stay. In this study several risk factors associated with postoperative delirium after cardiac surgery were identified. Utilizing the information may allow us to identifying patients at high risk of developing postoperative delirium prior to delirium onset.

Preoperative Risk Factors and Early Outcomes of Delirium in Valvular Open-Heart Surgery.

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Compared with coronary artery bypass grafting surgery, data regarding postoperative delirium are scant in valvular open-heart surgery. Therefore, the goal of this retrospective study was to investigate the incidence, preoperative risk factors, and early outcomes of delirium in a large group of patients undergoing valvular open-heart surgery. In 13,229 patients with isolated valvular or combined valvular and bypass surgery, the incidence of postoperative delirium was assessed until discharge. Independent risk factors of delirium were evaluated by multivariable logistic regression analysis. Moreover, we assessed the multivariable-adjusted risk of prolonged intensive care unit (ICU) stay (>48 hours) and in-hospital mortality in patients with delirium. Overall, the incidence of postoperative delirium was 8.4%. The incidence in patients experiencing a postoperative stroke or seizure was 23.1 and 29.7%, respectively. Twelve preoperative risk factors, mostly nonmodifiable, were independently associated with the risk of delirium, including advanced age, renal impairment, stroke, the need for emergency surgery, and severe preoperative anemia (hemoglobin < 9 g/dL). Postoperative delirium was associated with an adjusted odds ratio (OR) of prolonged ICU stay of 9.48 (95% confidence interval [CI]: 7.96-11.30). Adjusted in-hospital mortality was, however, significantly lower in patients with delirium versus patients without delirium (OR, 0.56; 95% CI: 0.38-0.83). In valvular open-heart surgery, postoperative delirium is a frequent neurological complication that is associated with other postoperative neurological complications and several, mostly nonmodifiable, preoperative risk factors. Although postoperative delirium was associated with a significantly increased risk of prolonged ICU stay, this did not translate into an increased short-term mortality.